Organic Chemistry Complete Sprint

15 Questions | Organic Chemistry | Total Marks: 60 | Total Time: 15 Minutes

Marking Scheme	+4 for correct, -1 for incorrect, 0 unattempted
Recommended Time	1 minute per question
Passing Marks	27 out of 60 (45%)

Question Paper

Q1. (Nomenclature) The IUPAC name of CH_3-CH_2-CHO is:

Marks: +4 / -1 | Time: 45 sec

A) Propanal
B) Propanol
C) Propanone
D) Ethanal

Q2. (Hybridization) The hybridization of carbon in ethyne ($HC\equiv CH$) is:

Marks: +4 / -1 | Time: 45 sec

A) sp^3
B) sp^2
C) sp
D) sp^3d

Q3. (Isomerism) Which type of isomerism is shown by ethanol and dimethyl ether (both C_2H_6O)?

Marks: +4 / -1 | Time: 60 sec

A) Chain isomerism
B) Functional isomerism
C) Position isomerism
D) Geometrical isomerism

Q4. (Reaction Mechanism) Markovnikov's rule is applied to which type of reaction?

Marks: +4 / -1 | Time: 60 sec

A) Addition of HX to unsymmetrical alkenes
B) Elimination of HX from alkyl halides
C) Substitution in aromatic rings
D) Oxidation of alcohols

Q5. (Functional Groups) The functional group present in carboxylic acids is:

Marks: +4 / -1 | Time: 30 sec

A) -CHO
B) -COOH
C) -OH

D) -CO-

Q6. (Reaction Types) Which reagent is used to convert an alkyl halide to an alcohol?

Marks: +4 / -1 | Time: 60 sec

A) Aqueous KOH
B) Alcoholic KOH
C) Na metal
D) Conc. H_2SO_4

Q7. (Aromatic Compounds) Benzene undergoes electrophilic substitution rather than addition mainly because:

Marks: +4 / -1 | Time: 60 sec

A) It has weak sigma bonds
B) Its aromatic ring stability is retained after substitution
C) It lacks pi electrons
D) It is highly reactive like alkenes

Q8. (Stereochemistry) A chiral carbon is one that is attached to:

Marks: +4 / -1 | Time: 45 sec

A) Four identical groups
B) Four different groups
C) Two identical and two different groups
D) Only hydrogen atoms

Q9. (Named Reactions) The Cannizzaro reaction is given by:

Marks: +4 / -1 | Time: 60 sec

A) Aldehydes with alpha-hydrogen
B) Aldehydes without alpha-hydrogen
C) Ketones with alpha-hydrogen
D) Carboxylic acids

Q10. (Polymers) Natural rubber is a polymer of:

Marks: +4 / -1 | Time: 45 sec

A) Ethylene
B) Isoprene
C) Styrene
D) Vinyl chloride

Q11. (Reaction Mechanism) SN1 reactions proceed through which type of intermediate?

Marks: +4 / -1 | Time: 45 sec

A) Free radical
B) Carbanion
C) Carbocation

D) Carbene

Q12. (Acidity/Basicity) Which of the following is the strongest acid?

Marks: +4 / -1 | Time: 60 sec

A) CH_3COOH
B) $ClCH_2COOH$
C) $Cl_2CHCOOH$
D) Cl_3CCOOH

Q13. (Carbonyl Chemistry) The reaction of an aldehyde with Tollens' reagent gives a positive test shown by:

Marks: +4 / -1 | Time: 45 sec

A) Formation of a silver mirror
B) Formation of a red precipitate
C) Evolution of a gas
D) Color change to blue

Q14. (Alcohols and Ethers) Which of the following tests distinguishes primary, secondary, and tertiary alcohols?

Marks: +4 / -1 | Time: 60 sec

A) Iodoform test
B) Lucas test
C) Tollens' test
D) Fehling's test

Q15. (Biomolecules Link) The hydrolysis of an ester in the presence of a base is called:

Marks: +4 / -1 | Time: 45 sec

A) Esterification
B) Saponification
C) Decarboxylation
D) Hydrogenation

Answer Key & Explanations

Q1. (Nomenclature) The IUPAC name of CH3-CH2-CHO is:

Correct Answer: A

Explanation: The compound has 3 carbons with a terminal -CHO group, making it an aldehyde named propanal.

Q2. (Hybridization) The hybridization of carbon in ethyne (HC≡CH) is:

Correct Answer: C

Explanation: Each carbon in ethyne forms a triple bond, using one sigma bond and two pi bonds, requiring sp hybridization.

Q3. (Isomerism) Which type of isomerism is shown by ethanol and dimethyl ether (both C2H6O)?

Correct Answer: B

Explanation: Ethanol (-OH group) and dimethyl ether (-O- group) have the same formula but different functional groups, showing functional isomerism.

Q4. (Reaction Mechanism) Markovnikov's rule is applied to which type of reaction?

Correct Answer: A

Explanation: Markovnikov's rule predicts that in addition of HX to an unsymmetrical alkene, H attaches to the carbon with more hydrogens already attached.

Q5. (Functional Groups) The functional group present in carboxylic acids is:

Correct Answer: B

Explanation: Carboxylic acids are characterized by the -COOH (carboxyl) functional group.

Q6. (Reaction Types) Which reagent is used to convert an alkyl halide to an alcohol?

Correct Answer: A

Explanation: Aqueous KOH promotes nucleophilic substitution (SN reaction), replacing the halide with -OH to form an alcohol. Alcoholic KOH instead favors elimination to form an alkene.

Q7. (Aromatic Compounds) Benzene undergoes electrophilic substitution rather than addition mainly because:

Correct Answer: B

Explanation: Substitution preserves the delocalized aromatic ring and its stability, while addition would destroy aromaticity, making substitution energetically favorable.

Q8. (Stereochemistry) A chiral carbon is one that is attached to:

Correct Answer: B

Explanation: A chiral (asymmetric) carbon has four different substituents attached, giving rise to non-superimposable mirror images (enantiomers).

Q9. (Named Reactions) The Cannizzaro reaction is given by:

Correct Answer: B

Explanation: Aldehydes lacking an alpha-hydrogen undergo self oxidation-reduction (Cannizzaro reaction) in the presence of concentrated base to form an alcohol and a carboxylate.

Q10. (Polymers) Natural rubber is a polymer of:

Correct Answer: B

Explanation: Natural rubber is cis-1,4-polyisoprene, a polymer formed from repeating isoprene (2-methyl-1,3-butadiene) units.

Q11. (Reaction Mechanism) SN1 reactions proceed through which type of intermediate?

Correct Answer: C

Explanation: SN1 reactions involve a two-step mechanism where the leaving group departs first, forming a carbocation intermediate before nucleophilic attack.

Q12. (Acidity/Basicity) Which of the following is the strongest acid?

Correct Answer: D

Explanation: Increasing number of electronegative chlorine atoms increases the -I (inductive) effect, stabilizing the conjugate base and increasing acidity; trichloroacetic acid is the strongest among these.

Q13. (Carbonyl Chemistry) The reaction of an aldehyde with Tollens' reagent gives a positive test shown by:

Correct Answer: A

Explanation: Tollens' reagent oxidizes aldehydes to carboxylic acids while Ag^+ ions are reduced to metallic silver, depositing as a silver mirror on the test tube.

Q14. (Alcohols and Ethers) Which of the following tests distinguishes primary, secondary, and tertiary alcohols?

Correct Answer: B

Explanation: The Lucas test uses $ZnCl_2$/HCl; tertiary alcohols react instantly (turbidity), secondary alcohols react within minutes, and primary alcohols show no visible reaction at room temperature.

Q15. (Biomolecules Link) The hydrolysis of an ester in the presence of a base is called:

Correct Answer: B

Explanation: Base-catalyzed hydrolysis of esters, producing an alcohol and a carboxylate salt (soap), is called saponification.